Level 3 Certificate
MATHEMATICAL STUDIES
1350/2A

Paper 2A Statistical Techniques

Mark scheme

June 2025

Version: 1.0 Final

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

No student should be disadvantaged on the basis of their gender identity and/or how they refer to the gender identity of others in their exam responses.

A consistent use of 'they/them' as a singular and pronouns beyond 'she/her' or 'he/him' will be credited in exam responses in line with existing mark scheme criteria.

Further copies of this mark scheme are available from aqa.org.uk

Glossary for Mark Schemes

Mathematical Studies examinations are marked in such a way as to award positive achievement wherever possible. Thus, for Mathematical Studies papers, marks are awarded under various categories.

If a student uses a method which is not explicitly covered by the mark scheme the same principles of marking should be applied. Credit should be given to any valid methods. Examiners should seek advice from their senior examiner if in any doubt.

M	Method marks are awarded for a correct method which could lead to a correct answer.
A	Accuracy marks are awarded when following on from a correct method. It is not necessary to always see the method. This can be implied.
B	Marks awarded independent of method.
ft	Follow through marks. Marks awarded for correct working following a mistake in an earlier step.
SC	Special case. Marks awarded for a common misinterpretation which has some mathematical worth.
M dep	A method mark dependent on a previous method mark being awarded.
B dep	A mark that can only be awarded if a previous independent mark has been awarded.
oe	Or equivalent. Accept answers that are equivalent. eg accept 0.5 as well as $\frac{1}{2}$
[a, b]	Accept values between a and b inclusive.
[a, b)	Accept values a ≤ value < b
3.14 …	Accept answers which begin 3.14 eg 3.14, 3.142, 3.1416
Use of brackets	It is not necessary to see the bracketed work to award the marks.

Examiners should consistently apply the following principles

Diagrams

Diagrams that have working on them should be treated like normal responses. If a diagram has been written on but the correct response is within the answer space, the work within the answer space should be marked. Working on diagrams that contradicts work within the answer space is not to be considered as choice but as working, and is not, therefore, penalised.

Responses which appear to come from incorrect methods

Whenever there is doubt as to whether a student has used an incorrect method to obtain an answer, as a general principle, the benefit of doubt must be given to the student. In cases where there is no doubt that the answer has come from incorrect working then the student should be penalised.

Questions which ask students to show working

Instructions on marking will be given but usually marks are not awarded to students who show no working.

Questions which do not ask students to show working

As a general principle, a correct response is awarded full marks.

Misread or miscopy

Students often copy values from a question incorrectly. If the examiner thinks that the student has made a genuine misread, then only the accuracy marks (A or B marks), up to a maximum of 2 marks are penalised. The method marks can still be awarded.

Further work

Once the correct answer has been seen, further working may be ignored unless it goes on to contradict the correct answer.

Choice

When a choice of answers and/or methods is given, mark each attempt. If both methods are valid then M marks can be awarded but any incorrect answer or method would result in marks being lost.

Work not replaced

Erased or crossed out work that is still legible should be marked.

Work replaced

Erased or crossed out work that has been replaced is not awarded marks.

Premature approximation

Rounding off too early can lead to inaccuracy in the final answer. This should be penalised by 1 mark unless instructed otherwise.

Continental notation

Accept a comma used instead of a decimal point (for example, in measurements or currency), provided that it is clear to the examiner that the student intended it to be a decimal point.

Q	Answer	Mark	Comments
1 (a)	$\frac{2}{5}$	B1	

<table>
<tr><th>Q</th><th>Answer</th><th>Mark</th><th>Comments</th></tr>
<tr><td rowspan="8">1 (b)</td><td>6 m 23 s – 2 m 18 s or 4 m 5 s
or
383 – 138 or 245 (s)
or
6.38(3…) – 2.3 or 4.08(3…)

or

6 m 23 s – 3 m 44 s or 2 m 39 s
or
383 – 224 or 159 (s)
or
6.38(3…) – 3.73(3…) or 2.65</td><td>M1</td><td>oe
range for Europe or Africa</td></tr>
<tr><td>4 m 5 s and 2 m 39 s and No
or
245 (s) and 159 (s) and No
or
4.08 (m) and 2.65 (m) and No</td><td>A1</td><td>oe
must compare consistent units</td></tr>
<tr><td colspan="3">Additional Guidance</td></tr>
<tr><td colspan="2">Condone other representation of time for method and accuracy eg 4:05 or 4.05 or 405</td><td></td></tr>
<tr><td colspan="2">Ignore further working for comparison eg doubling the range for Africa</td><td></td></tr>
<tr><td colspan="2">No may be implied eg Jaydon is wrong</td><td></td></tr>
<tr><td colspan="2">6.23 – 2.18 = 4.05 and 6.23 – 3.44 = 2.79 and No</td><td>M1 A0</td></tr>
<tr><td colspan="2">623 – 218 = 405 and 623 – 344 = 279 and No</td><td>M1 A0</td></tr>
</table>

<table>
<tr><th>Q</th><th>Answer</th><th>Mark</th><th colspan="2">Comments</th></tr>
<tr><td rowspan="6">1 (c)</td><td>Valid explanation
eg
He has included the 2026 eclipse (2 m 18 s) twice
or
He has only worked out the median/ used the times for North America (2 m 18 s and 2 m 36 s)
or
He did not include 2027 eclipse (in Europe, 6 m 23 s)
or
He found the value between the first and second value from the three values
or
He worked out the mean for North America (2 m 18 s and 2 m 36 s)</td><td>B1</td><td colspan="2">oe</td></tr>
<tr><td>2 m 36 s</td><td>B1</td><td colspan="2">correct median must be clearly stated or on the answer line</td></tr>
<tr><td colspan="4">Additional Guidance</td></tr>
<tr><td colspan="3">He should have just found the median of 2 m 18 s, 2 m 36 s, 6 m 23 s and correct median is 2 m 36 s</td><td>B0 B1</td></tr>
<tr><td colspan="3">He worked out the mean (not the median)</td><td>B0</td></tr>
<tr><td colspan="3">He found the median/mean of 2 m 18 s and 2 m 36 s (not all three values)</td><td>B1</td></tr>
</table>

Q	Answer	Mark	Comments
2 (a)	**Alternative method 1: broadcasting as a proportion of total revenue**		
	294 ÷ 731 or 0.4(0…) or 74 ÷ 190 or 0.38(9…) or 0.39	M1	condone reciprocals
	0.4(0…) **and** 0.38(9..) or 0.39 **and** yes or 2.48(6…) or 2.49 or 2.5 **and** 2.56(7…) or 2.57 or 2.6 **and** yes	A2	oe percentages A1 0.4(0…) **and** 0.38(9...) or 0.39 or 2.48(6…) or 2.49 or 2.5 **and** 2.56(7…) or 2.57 or 2.6
	Alternative method 2: recalculating AFC Ajax proportion		
	294 ÷ 731 (× 190)	M1	calculating AFC Ajax broadcast revenue based on Manchester City proportion
	76.(4…) (€ million) and yes	A2	A1 76.(4…) (€ million)
	Alternative method 3: recalculating Manchester City proportion		
	74 ÷ 190 (× 731)	M1	calculating Manchester City broadcast revenue based on AFC Ajax proportion
	284.7(0…) or 285 (€ million) and yes	A2	A1 284.7(0…) or 285 (€ million)

The mark scheme for Question 2 (a) continues on the next page

<table>
<tr><th>Q</th><th>Answer</th><th>Mark</th><th>Comments</th></tr>
<tr><td rowspan="8">2 (a) cont'd</td><td colspan="3">Alternative method 4: Scale factor of broadcast compared to scale factor of totals</td></tr>
<tr><td>294 ÷ 74 or 3.9(7….)
or
731 ÷ 190 or 3.8(4…)</td><td>M1</td><td>condone reciprocals</td></tr>
<tr><td>3.9(7…) or 4
and
3.8(4…)
and yes

or

0.25(1…) or 0.3
and
0.25(9…) or 0.3
and yes</td><td>A2</td><td>A1
3.9(7…) or 4
and
3.8(4…)

or

0.25(1…) or 0.3
and
0.25(9…) or 0.3</td></tr>
<tr><td colspan="3">Additional Guidance</td></tr>
<tr><td colspan="2">Other comparable proportions may be used
eg 294 ÷ 437 = 0.67 and 74 ÷ 116 = 0.64 and yes</td><td></td></tr>
<tr><td colspan="2">Conclusion should support statement</td><td></td></tr>
<tr><td colspan="2">Allow rounding or truncation if method seen</td><td></td></tr>
<tr><td colspan="2">Comparison of non-broadcast revenue</td><td>M0 A0</td></tr>
</table>

Q	Answer	Mark	Comments
2 (b)	**Alternative method 1**		
	AC Milan	B1	
	104 ÷ 33 or 3.1(51..) or 3.2 or 133 ÷ 33 or 4.(03)	M1	
	3.1(51..) or 3.2 and 4(.03)	A1	matchday unit value of 1 implied by these values SC2 Real Madrid and 3.3(40...) and 3.5(16...)
	Alternative method 2		
	AC Milan	B1	
	33 × 3 or 99 or 33 × 4 or 132	M1	
	99 and 132	A1	SC2 Real Madrid and 273 and 364
	Alternative method 3		
	AC Milan	B1	
	270 ÷ 8 or (104 + 133 + 33) ÷ 8 or 33.75 or 33.8 or 34	M1	
	[101, 102] and [135,136] and [33.75,34] or [3.05, 3.09] and [3.9, 3.95] and [0.97, 0.98]	A1	SC2 Real Madrid and 268.125 and 357.5 and 89.375 or Real Madrid and 267.75 and 357 and 89.25

The mark scheme for Question 2 (b) continues on the next page

	Answer	Mark	Comments
2 (b) cont'd	**Alternative method 4**		
	AC Milan	B1	
	104 ÷ 3 or 34.(6…) or 133 ÷ 4 or 33.(25)	M1	
	34.(6…) and 33.(25)	A1	SC2 Real Madrid and 101.(3…) and 80
	Alternative method 5		
	AC Milan	B1	
	104 – 2 x (133-104) or 104 – 2 x 29	M1	
	46 with M1 scored	A1	SC2 Real Madrid and 304 – 2 x (320-304) = 288
	Alternative method 6		
	AC Milan	B1	
	104 ÷ 270 or 0.38(5…) or 0.39 or 133 ÷ 270 or 0.49(…) or 33 ÷ 270 or 0.12(….)	M1	
	[3.16, 3.2] and [4, 4.11] and 1 or 0.38(5…) or 0.39 and 0.375 and 0.49(…) and 0.5 and 0.12(….) and 0.125	A1	SC2 Real Madrid and [3.38, 3.52] and [3.2, 3.4] **or** Real Madrid and 0.44(5…) or 0.45 and 0.42(5…) or 0.43 and 0.12(7…) or 0.13 and 0.375 and 0.5 and 0.125

The mark scheme for Question 2 (b) continues on the next page

	Additional Guidance	
2 (b) cont'd	Allow any equivalent method eg $\frac{4}{3} \times 104 = 138.7$ and $\frac{1}{3} \times 104 = 34.7$	

Q	Answer	Mark	Comments
2 (c)	three values added in domains [303.5, 304.5) + [319.5, 320.5) + [90.5, 91.5) that give answer in range [713.5, 714.5)	B1	
	Additional Guidance		
	Final answer not required but if seen must be correct or shown as 714		
	303.5 + 319.5 + 90.5		B1
	304.5 + 319.5 + 90.5		B0
	303.4 + 320 + 91		B0

Q	Answer	Mark	Comments
2 (d)	270 ÷ 1.16 or 233 or 232.(7…)	M1	oe
	144 ÷ 0.94 or 153.(19…) or 153.2	M1	oe
	80 with M2 scored	A1	condone values stated in millions
	Additional Guidance		
	153 from incorrect method eg 144 × 1.06 = 152.64 = 153		M0
	153 and 233 and 80 with no incorrect method		M1 M1 A1

<table>
<tr><th>Q</th><th>Answer</th><th>Mark</th><th colspan="2">Comments</th></tr>
<tr><td rowspan="9">2 (e)</td><td>Any two valid criticisms from distinct categories

Shape
Pyramid / 3D
Does not start at zero
Shape is above and below the horizontal axis
Shape does not align with scale
Shapes start from different points

Axis label
No y-axis label
No x-axis label
Angle of labels
No units eg currency or millions

Scale / Gridlines
Gridlines only every 50 units
Negatives signs are on the gridlines
The gaps on the y-scale are too large
The vertical axis only extends to 50 (so the 49 label is outside the graph area)</td><td>B2</td><td colspan="2">B1 one valid criticism
allow improvements to imply criticism
two criticisms may be in one response
ignore any extra criticisms</td></tr>
<tr><td colspan="4">Additional Guidance</td></tr>
<tr><td colspan="3">-25 is too close to -50 on the scale (Shape category)</td><td>B1</td></tr>
<tr><td colspan="3">Should have used rectangles (Shape category)</td><td>B1</td></tr>
<tr><td colspan="3">More gridlines (Gridlines category)</td><td>B1</td></tr>
<tr><td colspan="3">Should extend the vertical axis beyond 50 (Scale / Gridlines category)</td><td>B1</td></tr>
<tr><td colspan="3">Inaccurate</td><td>B0</td></tr>
<tr><td colspan="3">No vertical gridlines</td><td>B0</td></tr>
<tr><td colspan="3">No gridlines</td><td>B0</td></tr>
</table>

Q	Answer	Mark	Comments
2 (f) (i)	(49 + 21 + 13 + 9 – 24 – 25 – 69 – 369) ÷ 8 or (–) 395 ÷ 8	M1	
	– 49.375 with M1 scored or loss of 49.375 with M1 scored	A1	allow rounding or truncating SC1 – 400 ÷ 8 = –50 or (–)49.3(75) or (–)49.38 or (–)49.4 without working
	Additional Guidance		
	Ignore units		
	Calculating the median or other measures does not support the statement		
	49 + 21 + 13 + 9 – 24 – 25 – 69 – 369 ÷ 8		M0 unless recovered
	SC1 awarded for rounding (–)395 to (–)400 prior to division		
	The mean is (–)49 without working		M0 A0

Q	Answer	Mark	Comments
2 (f) (ii)	One valid reason eg The data is old / only from one year/season The clubs are not representative of current European champions The clubs are not representative of champions from other European leagues Only used a small sample of clubs PSG's loss has skewed the mean average / PSG is an outlier The median average is a loss of €7.5m Without PSG the mean loss is only €3.7 m Data affected by Covid	B1	
	Additional Guidance		
	The majority of clubs do not make a loss of more than €50m		B1
	Half of the clubs made a profit		B0
	Only represents clubs who win their league		B0
	There was an outlier		B0
	None of the clubs are close to losing the mean amount		B0

Q	Answer	Mark	Comments
3 (a)	$17 \times 39\,000$ and $8 \times 32\,000$ and $15 \times 35\,000$ or 663 000 and 256 000 and 525 000	M1	
	$17 + 8 + 15$ or 40	M1	
	$\dfrac{17\times39000+8\times32000+15\times35000}{17+8+15}=36100$ or $\dfrac{1444000}{40} = 36\,100$ with M2 seen	A1	
	Additional Guidance		
	Allow figures in thousands for the first M1 eg 17×39 and 8×32 and 15×35 or 663 and 256 and 525		

Q	Answer	Mark	Comments
3 (b)	Correct population statement using '**senior nurses'** and '**London**' eg (all) senior nurses in London or number of senior nurses in London	E1	condone 'the salaries of (all) senior nurses in London' do not allow 'all nurses in London'

Q	Answer	Mark	Comments
3 (c)	Correct explanation eg increase the sample size/number of people or get data from more hospitals (in London) or use random samples	E1	oe

Q	Answer	Mark	Comments
4	N(0, 1)	B1	

Q	Answer	Mark	Comments
5 (a)	92% value → (±) [1.75, 1.751]	B1	implied by correct answer in range condone use of 1.8
	$410 \pm \text{their } [1.75, 1.751] \times \sqrt{\frac{121}{100}}$ or 410 ± their [1.75, 1.751] × 1.1 or 410 ± 1.92(…) or 410 ± 1.93	M2	M1 for one error in the equation eg no √ sign for 121 or 100 or fraction reversed × $\sqrt{\frac{100}{121}}$ their [1.75, 1.751] does not count as an error if it is in the range (0, 4]
	([408, 408.1], [411.9, 412])	A1	allow [411.9, 412] and [408, 408.1]
	425 lies outside the (92% confidence) interval and claim is incorrect or unlikely or not valid	E2ft	oe ft their confidence interval E1 425 lies outside the (92% confidence) interval condone ‘it’ for ‘425’ condone ‘range’ or ‘limit’ for ‘interval’
	Additional Guidance		
	For 92% value, condone use of 1.8 throughout		
	If candidates use 121 or 100 instead of √121 or 11 or √100 or 10 can score B1M1A0 If both 121 and 100 are used instead of √121 and √100 can score B1M0A0		
	Not using ± and omitting either + or – in the equation counts as one error		
	([408, 408.1], [411.9, 412]) seen without method scores B1M2A1		
	(0, 4] → 0 < value ⩽ 4		
	No confidence interval calculated or stated scores E0		

Q	Answer	Mark	Comments
5 (b)	$1 - 0.92$ or 0.08 or 4% or 0.04 seen	M1	oe fraction or percentage
	0.04	A1	oe fraction or percentage
	Additional Guidance		
	Ignore conversion between decimals and percentages after correct answer seen Other further work after 0.04 seen scores M1A0		

Q	Answer	Mark	Comments
6	1	B1	

Q	Answer	Mark	Comments
7 (a) (i)	61	B1	

Q	Answer	Mark	Comments
7 (a) (ii)	$-0.92(8\ldots)$ or -0.93	B1	
	Strong negative	E1ft	ft their pmcc only
	Additional Guidance		
	No pmcc written scores B0E0		
	‘strong positive’ for $0.5 \leqslant$ pmcc < 1 or ‘strong negative’ for $-1 <$ pmcc $\leqslant -0.5$ Condone ‘weak positive’ for $0 <$ pmcc $\leqslant 0.5$ or ‘weak negative’ for $-0.5 \leqslant$ pmcc < 0		
	Accept ‘moderate positive’ for $0.3 \leqslant$ pmcc $\leqslant 0.7$ or ‘moderate negative’ $-0.7 \leqslant$ pmcc $\leqslant -0.3$		

Q	Answer	Mark	Comments
7 (a) (iii)	pmcc gets closer to 0 or less negative	E1ft	oe ft their pmcc from part **a(ii)**
	Type stays the same (negative) and strength becomes weak(er)	E1ft	oe ft their type and strength from part **a(ii)**

Additional Guidance

If new pmcc calculated –0.77(…) and correctly compared with their negative pmcc from part **(a)(ii)** then scores the first E1

Example	First E mark (**pmcc**)
From **(a)(ii)** if their **pmcc** is negative value	Increases Becomes less negative Gets closer to 0 Is greater Becomes higher Goes up Raises / rises Gets smaller Condone eg 'Decrease closer to 0'
From **(a)(ii)** if their **pmcc** is positive value	Decreases Becomes less positive Gets closer to 0 Is less Becomes lower Goes down Drops / falls Gets smaller

Example	Second E mark (**type and strength**)
From **(a)(ii)** their **Type** is negative **Strength** is strong or moderate	**Type** Stays the same oe Stays negative **Strength** Becomes weak(er) oe Becomes less strong
From **(a)(ii)** their **Type** is positive **Strength** is strong or moderate	**Type** Stays the same oe Stays positive **Strength** Becomes weak(er) oe Becomes less strong

<table>
<tr><th>Q</th><th>Answer</th><th>Mark</th><th>Comments</th></tr>
<tr><td rowspan="8">7 (a) (iv)</td><td>Other factors affect amount of time spent
or
It’s a small sample
or
Sample is not representative (of the whole population)</td><td>E1</td><td>allow a counter example using two residents from data or in context
condone correlation does not imply causation</td></tr>
<tr><td colspan="3">Additional Guidance</td></tr>
<tr><td colspan="3">An older resident may have to use the internet for their job scores E1
A resident aged 45 is spending more time on internet than a resident aged 42 scores E1</td></tr>
<tr><td colspan="3">The oldest person is only 67 so can’t comment on those older than 67 scores E0</td></tr>
<tr><td colspan="3">Reference to the outlier (61, 23) alone without comparison with other data scores E0</td></tr>
<tr><td colspan="3">It is not within the data range scores E0</td></tr>
<tr><td colspan="3">Not a perfect correlation scores E0</td></tr>
<tr><td colspan="3">Only stating that some data does not follow the trend scores E0</td></tr>
</table>

Q	Answer	Mark	Comments
7 (b) (i)	(y =) [40, 40.4] – [0.538, 0.54] x	B2	implied by substitution of $x = 70$ into the correct equation or expression B1 for [40, 40.4] or –[0.538, 0.54]
	Substitutes $x = 70$ into any linear equation or expression in the form of $a + bx$ or Uses their equation to draw their regression line correctly and attempts to find the value of y when $x = 70$	M1	
	[2.2, 2.74] or 2 or 3	A1ft	ft their stated equation of the regression line
	Additional Guidance		
	[2.2, 2.74] or 2 or 3 must come from using the correct equation or expression		
	For M1, if their regression line is drawn, it must go through two correctly plotted points and extended to at least $x = 70$		

Q	Answer	Mark	Comments
7 (b) (ii)	It involves extrapolation or It is not within the data range or Trends may change outside the range of the given data or That resident might be an outlier too	E1	oe
	Additional Guidance		
	There may be other outliers scores E0		

Q	Answer	Mark	Comments
8 (a)	0.5	B1	oe

Q	Answer	Mark	Comments
8 (b)	$(z =)\dfrac{950-1200}{150}$ or –1.66(6…) or –1.67	M1	implied by [0.047, 0.0485] or [0.9515, 0.953] condone $\dfrac{1200-950}{150}$ or 1.66(6…) or 1.67 condone –1.7 or 1.7 for method
	[0.047, 0.0485] or [0.9515, 0.953]	A1	oe implied by correct answer ignore subsequent rounding if answer in range seen
	$(z =)\dfrac{1250-1200}{150}$ or 0.33(3…)	M1	implied by [0.629, 0.631] or [0.369, 0.371] condone $\dfrac{1200-1250}{150}$ or –0.33(3…) condone 0.3 or –0.3 for method
	[0.629, 0.631] or [0.369, 0.371]	A1	oe implied by correct answer ignore subsequent rounding if answer in range seen
	their [0.629, 0.631] – their [0.047, 0.0485] or their [0.629, 0.631] – (1 – their [0.9515, 0.953]) or 1 – their [0.047, 0.0485] – their [0.369, 0.371] or [0.9515, 0.953] – their [0.369, 0.371] or their [0.629, 0.631] + their [0.9515, 0.953] – 1	M1dep	oe implied by correct answer dep on M2
	[0.58, 0.584]	A1	oe ignore subsequent rounding if answer in range seen
	Additional Guidance		
	[0.58, 0.584] without working or method or contradiction scores full marks		
	Accept equivalent fractions throughout		

Q	Answer	Mark	Comments
8 (c)	(±) 0.67(45) or (±) 0.675	B1	implied by [1098, 1100] or [1300, 1302]
	$\dfrac{Q - 1200}{150} = (\pm)\ 0.67(45)$	M1	oe equation implied by [1098, 1100] or [1300, 1302] allow [–4, 4] except ±0.25 and ±0.75 for 0.67(45)
	(Q_1 =) 1200 – their 0.67(45) × 150 or [1098, 1100] or (Q_3 =) 1200 + their 0.67(45) × 150 or [1300, 1302]	M1	oe implied by correct answer
	[1098, 1100] **and** [1300, 1302]	A1	oe implied by correct answer
	[200, 204]	A1	
	Additional Guidance		
	[1098, 1100] or [1300, 1302] without working or not from incorrect working scores B1M1M1		
	Allow other letters for Q throughout		

Q	Answer	Mark	Comments
8 (d)	Approximately 95% of observations lie within 2 standard deviations of the mean or Approximately 99.(…) % or 100% of observations lie within 3 standard deviations of the mean or States $P(900 < R < 1500) = 0.95(\ldots)$	E1	oe
	95% $\neq$ (is not equal to) 100% or Four standard deviations $\neq$ (is not equal to) 100% or Within two standard deviations $\neq$ (is not equal to) 100% or States that it should be more than 4 standard deviations	E1	oe
	Additional Guidance		
	Observations that lie within 2 standard deviations of the mean do not represent the whole population scores E0E1		
	There are data outside 2 standard deviations from the mean scores E0E1		